Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cylinders and spheres, and volume. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	5
Topic	Three dimensional shapes II
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume
Theme	—
Curriculum objectives	• identify properties of cylinders and spheres • find the volume of a cube, a cuboid and a prism
Previous knowledge	• properties of cubes and cuboid • properties of pyramids and cones
Lesson sequence	• identify properties of cylinders and spheres — the teacher displays shapes of cylinders and spheres; the student identify the properties; the teacher draws the shapes on the board and adds to the properties already mentioned by the students. Set for the students: 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces. • find the volume of a cube, a cuboid and a prism — the teacher explains how to derive the formula for volume of cube and cuboid, and works the volume of a prism as area of base times height; the student listen and ask questions where they do not understand; the student copy worked examples into their notes
Instructional materials	• shapes of cylinders and spheres.
Evaluation	• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high. • Calculate the volume of cuboid with dimensions 5m by 2m by 3m. • What is the volume of a cube of 4cm × 4cm × 4cm. • A rectangular tank hold 6m long by 2m wide holds $36m^3$ of water. How deep is the water in the tank.

Lesson objectives

1.	identify properties of cylinders and spheres
2.	find the volume of a cube, a cuboid and a prism

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cylinders and spheres, and volume

Class: JSS1 **Subject:** Mathematics **Week:** 5

Section 1: Cylinders and Spheres, and Finding Volume

Solid shapes are all around you — tins, balls, boxes, and bricks. In this section you will get to know two of them well (the cylinder and the sphere), and then find the volume of three solid shapes: the cube, the cuboid and the prism.

Cylinders
A cylinder looks like a tin of tomatoes or a drinking can. Its key properties are:
• It has two flat circular faces — one at the top and one at the bottom — and they are the same size.
• It has one curved surface that joins the two circles.
• If you cut it straight across, the slice is always a circle the same size (its cross-section is a constant circle).
• It has no vertices (corners).
• Its height is the distance between the two circular faces.

Spheres
A sphere looks like a ball — a football or a marble. Its key properties are:
• It is perfectly round.
• Every point on its surface is the same distance from its centre. That distance is called the radius.
• It has no flat faces, no edges and no vertices at all.
• No matter how you slice it, a cut gives you a circle.

Volume
Volume measures how much space a solid takes up. You work it out by multiplying a shape's length, width and height, and you answer in cubic units (like cm³, meaning centimetres cubed).

• Cube — all its edges are equal. Volume = side × side × side, or side³.
• Cuboid — a box with three different lengths. Volume = length × width × height.
• Prism — a shape with a cross-section that is the same all the way through (for example, a triangular prism). Volume = area of the cross-section × length.

Worked Example 1 — the volume of a cube
Find the volume of a cube whose side is 4 cm.

Volume of a cube = side × side × side
= 4 × 4 × 4
= 64 cm³

So the cube takes up 64 cubic centimetres of space.

Worked Example 2 — the volume of a cuboid
A cuboid is 5 cm long, 3 cm wide and 2 cm high. Find its volume.

Volume of a cuboid = length × width × height
= 5 × 3 × 2
= 30 cm³

The order of the three measurements does not matter — you always multiply them together.

Worked Example 3 — the volume of a prism
A triangular prism has a triangular cross-section with area 12 cm², and its length is 6 cm. Find its volume.

Volume of a prism = area of cross-section × length
= 12 × 6
= 72 cm³

The prism holds 72 cubic centimetres.

Practice

1. A drinks tin is in the shape of a cylinder. How many flat faces does it have, and what shape are they?
2. Which solid shape has no flat faces, no edges and no vertices? Name one thing in your home that has this shape.
3. Find the volume of a cube whose side is 5 cm.
4. A cuboid is 8 cm long, 4 cm wide and 3 cm high. Find its volume.
5. A triangular prism has a triangular cross-section of area 15 cm² and a length of 4 cm. Find its volume.

Section 2: Cylinders, Spheres, and Finding Volume

Before we measure shapes, let's make sure we know them. A cylinder is like a tin or a drum: it has two flat, circular faces (top and bottom) that are the same size and face each other, and one curved surface that joins them. Its two circular faces are identical and lie flat and parallel to each other. A sphere is perfectly round like a ball or an orange: it has just one surface, the whole outside is curved, and there is no flat face, no edge and no corner. From the centre of a sphere, every point on its surface is exactly the same

distance away.
Now for the main job of this section: finding volume. The volume of a solid is the amount of space it takes up, and we measure it in cubic units such as cm³ or m³. For a box-shaped solid, the volume is found by multiplying its three measurements together.
A cube is a box where every edge is the same length. If one edge is length s, then:
Volume of a cube = s × s × s = s³
A cuboid is a box where the faces are rectangles, and its three edges can all be different lengths — length l, breadth b and height h. Then:
Volume of a cuboid = l × b × h
Many solids are prisms. A prism is a solid with the same cross-section all the way through — think of a Toblerone bar, a pipe, or a wedge of cheese. For any prism:
Volume of a prism = area of cross-section × length
So once you can work out the area of the shape at the end, you multiply it by how long the prism is.

Example 1 — Volume of a cube. A cube has edges of length 4 cm. Find its volume.
Volume = 4 × 4 × 4 = 64. So the volume is 64 cm³.
Example 2 — Volume of a cuboid. A box measures 5 cm long, 3 cm broad and 2 cm high. Find its volume.
Volume = 5 × 3 × 2 = 30. So the volume is 30 cm³.
Example 3 — Volume of a prism. A prism has a triangular end with an area of 8 cm², and it is 6 cm long. Find its volume.
Volume = area of cross-section × length = 8 × 6 = 48. So the volume is 48 cm³.
Notice the pattern: a cube and a cuboid are actually special prisms, because their cross-section is the same all the way through. For a cuboid, that cross-section is a rectangle, so its area is length × breadth, and multiplying by the height again gives length × breadth × height.

Practice

1. Look at these two solids. Which one is a cylinder and which one is a sphere?
   (a) A solid with two identical flat circular faces and one curved surface.
   (b) A perfectly round solid with only a curved surface and no flat faces.
2. A cube has edges of length 6 cm. Find its volume.
3. A cuboid is 7 cm long, 4 cm broad and 3 cm high. Find its volume.
4. A prism has a triangular cross-section with an area of 5 cm² and it is 9 cm long. Find its volume.
5. A rectangular box is 10 cm long, 6 cm broad and 5 cm high. How many cubes of edge 1 cm can fit inside it?

Check your work

Attempt every question before you read this page.

Cylinders and Spheres, and Finding Volume

1. It has two flat faces, and both are circles.

   A cylinder has two circular faces (one on top and one at the bottom) plus one curved surface joining them. So it has two flat faces, and they are circles.
2. A sphere. One example is a ball or an orange.

   A sphere is perfectly round: every point on its surface is the same distance from its centre, and it has no flat faces, edges or vertices. Any ball, marble or orange is a sphere.
3. 125 cm³

   Volume of a cube = side × side × side = 5 × 5 × 5 = 125 cm³.
4. 96 cm³

   Volume of a cuboid = length × width × height = 8 × 4 × 3 = 96 cm³.

5. 60 cm³

   Volume of a prism = area of cross-section × length = 15 × 4 = 60 cm³.

Cylinders, Spheres, and Finding Volume

1. (a) is a cylinder; (b) is a sphere.
2. 6 × 6 × 6 = 216 cm³

   For a cube, volume = s × s × s. Here s = 6, so volume = 6 × 6 × 6 = 216. The answer is 216 cm³.
3. 7 × 4 × 3 = 84 cm³

   For a cuboid, volume = length × breadth × height = 7 × 4 × 3 = 84. The answer is 84 cm³.
4. 5 × 9 = 45 cm³

   For a prism, volume = area of cross-section × length = 5 × 9 = 45. The answer is 45 cm³.
5. Its volume is 10 × 6 × 5 = 300 cm³, so 300 one-cubic-centimetre cubes fit inside.

   Each 1 cm cube takes up 1 cm³. The volume of the box is length × breadth × height = 10 × 6 × 5 = 300 cm³, so 300 of the small cubes fit inside.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-a47056f0	**Date**	
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____